

Campus Est, Via La Santa 1,
CH-6962 Lugano-Viganello
T +41 (0)58 666 66 66, F +41 (0)58 666 66 61
info@idsia.ch, www.idsia.ch
N. IVA: CHE-108.955.570 IVA

Deep Learning for System Identification

Date: From June 15 to June 19, 2026

Location: KU Leuven, Leuven, Belgium

Organizer: EECI - European Embedded Control Institute

Lecturers: Dario Piga, Marco Forgione, e-mails: dario.piga@supsi.ch, marco.forgione@supsi.ch

Summary of the Course

This course introduces students to deep learning tools for nonlinear system identification. The topics covered include:

- Introduction to system identification
- Linear-in-the-parameter models and LASSO
- Introduction to deep learning
- Feedforward network, RNN, and LSTM for identification of dynamical systems
- Neural state-space models
- Neural networks with LTI blocks: dynoNet and beyond
- Meta learning for system identification

In the exercise sessions, students will apply the methodologies presented in the preceding lesson. These sessions will be conducted in Python, using the PyTorch deep learning framework. An introduction to PyTorch will be provided during the course.

Prerequisites

- Basic knowledge of dynamical systems
- Basic knowledge of Python. We encourage conducting exercise sessions in groups (max. 3 persons), with at least one member of the group having basic knowledge of PyTorch.
- We recommend installing all the required software packages before the start of the course.

Final exam is possible to receive a Certificate of Accreditation of 3 ECTS credits upon completion of an oral (remote) exam. If interested, please inform us at the beginning of the course.

Tentative Schedule

Day 1 (Monday, June 15)

Time	Title/Topic	Lecturer
14:00-15:30	Introduction to SYSID	D. Piga
15:30-16:00	Break	
16:00-17:30	Introduction to SYSID	D. Piga

Day 2 (Tuesday, June 16)

Time	Title/Topic	Lecturer
09:00-10:30	Deep Learning	D. Piga
10:30-11:00	Break	
11:00-12:30	PyTorch – Low-level commands	M. Forgiione
12:30-14:00	Lunch break	
14:00-15:30	PyTorch – Higher-level abstractions	M. Forgiione
15:30-16:00	Break	
16:00-17:30	Exercise (Regression in PyTorch)	M. Forgiione

Day 3 (Wednesday, June 17)

Time	Title/Topic	Lecturer
09:00-10:30	PyTorch – System Identification	M. Forgiione
10:30-11:00	Break	
11:00-12:30	Exercise (NARX/RNN)	D. Piga

Day 4 (Thursday, June 18)

Time	Title/Topic	Lecturer
09:00-10:30	Neural State Space	M. Forgiione
10:30-11:00	Break	
11:00-12:30	Exercise (Neural State Space)	M. Forgiione
12:30-14:00	Lunch break	
14:00-15:30	dynoNet	D. Piga
15:30-16:00	Break	
16:00-17:30	Exercise (dynoNet)	D. Piga

Day 5 (Friday, June 19)

Time	Title/Topic	Lecturer
09:00-10:30	Meta learning for SYSID	M. Forgiione
10:30-11:00	Break	
11:00-12:30	Exercise (Meta learning static regression)	M. Forgiione